

PHY201: Waves & Optics

Problem Set 6

September 30, 2025

1. The equation of a transverse wave traveling along a string is given by $y = 0.3 \sin \pi(0.5x - 50t)$, where y and x are in centimeters and t is in seconds.
 - (a) Find the amplitude, wavelength, wave number, frequency, period, and velocity of the wave.
 - (b) Find the maximum transverse speed of any particle in the string.
2. A wave of frequency 20 sec^{-1} has a velocity of 80 m/sec .
 - (a) How far apart are two points whose displacements are 30° apart in phase?
 - (b) At a given point, what is the phase difference between two displacements occurring at times separated by 0.01 sec ?
3. It is observed that a pulse requires 0.1 sec to travel from one end to the other of a long string. The tension in the string is provided by passing the string over a pulley to a weight which has 100 times the mass of the string.
 - (a) What is the length of the string?
 - (b) What is the equation of the third normal mode?
4. The following two waves in a medium are superposed:

$$y_1 = A \sin(5x - 10t); \quad y_2 = A \sin(4x - 9t)$$

where x is in meters and t in seconds.

- (a) Write an equation for the combined disturbance.
 - (b) What is its group velocity?
 - (c) What is the distance between points of zero amplitude in the combined disturbance?
5. Consider a longitudinal sinusoidal wave $\xi = \xi_0 \cos 2\pi k(x - vt)$ traveling down a rod of mass density ρ , cross-sectional area S , and Young's modulus Y . Show that if the stress in the rod is due solely to the presence of the wave, the kinetic-energy density is $\frac{1}{2}\rho S(\partial\xi/\partial t)^2$ and the potential-energy density is $\frac{1}{2}YS(\partial\xi/\partial x)^2$. Therefore show that the kinetic energy per wavelength and the potential energy per wavelength both equal $\frac{1}{4}(\rho S\lambda)u_0^2$, where u_0 is the maximum particle velocity ($\partial\xi/\partial t$).